

POWER BI – WEEK 1 REPORT

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ROLL NO : E25DA040

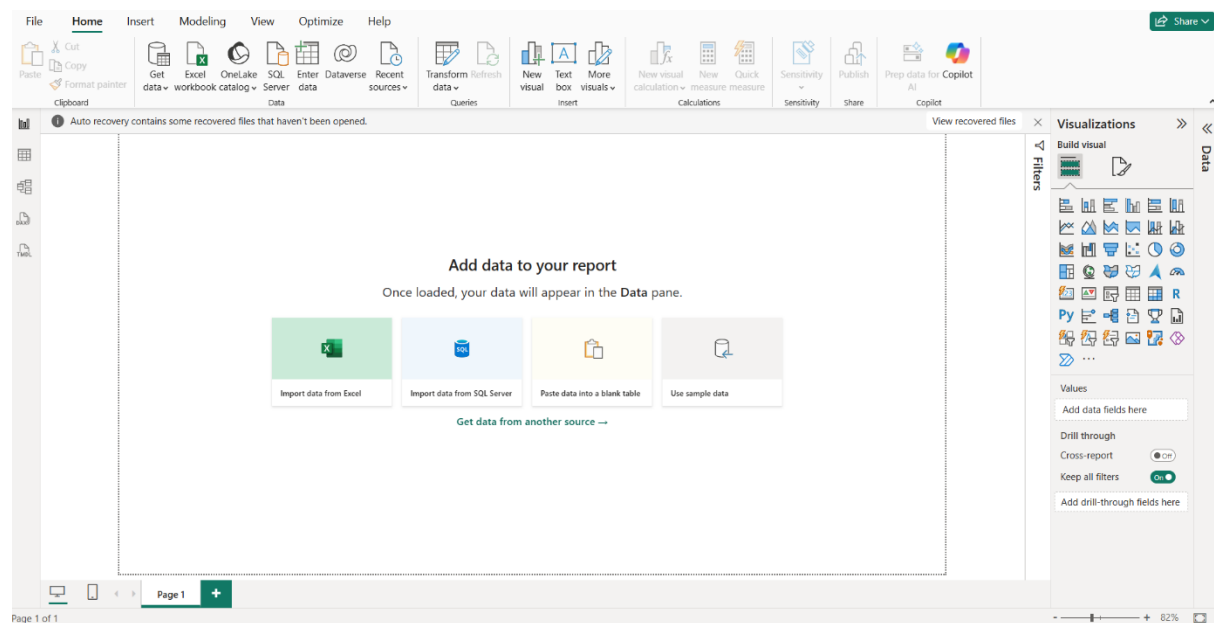
POWER BI

STEP : 1

Open Power BI

Microsoft Power BI Desktop was opened to begin the data analysis process.

Screenshot:

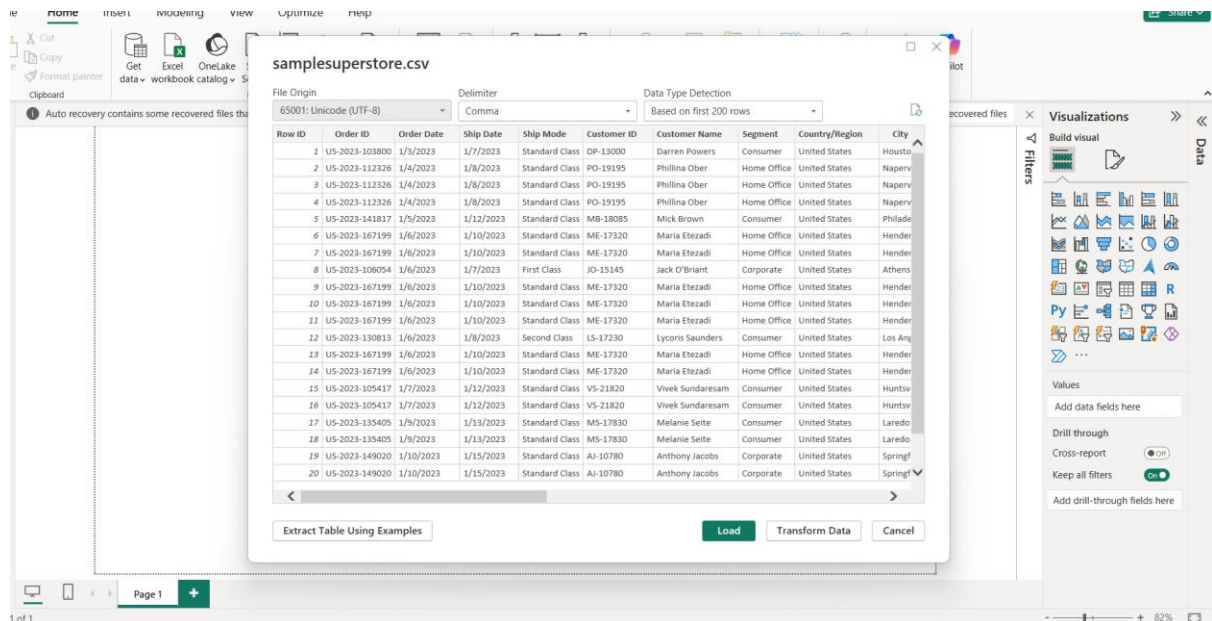


STEP : 2

Load the Dataset and Click Transform Data

The Superstore dataset was imported into Power BI using the Text/CSV option. After selecting the dataset, the Transform Data option was selected to open the data in Power Query Editor.

Screenshot:



STEP : 3

Change Order Date Using Locale

The Order Date column was converted into the appropriate date format using the Using Locale option.

Steps:

Transform → Data Type → Using Locale → Date → Select Locale → OK

Screenshot:

Queries [1]

samplesuperstore

Row ID Order ID Order Date Ship Date Ship Mode

Valid 100% Valid 100% 1.2 Decimal Number Valid 100% Valid 100% Valid 100%

Error 0% Error 0% \$ Fixed decimal number Error 0% Error 0% Error 0%

Empty 0% Empty 0% Whole Number Empty 0% Empty 0% Empty 0%

1 1 US-2023-103800 % Percentage 2/2023 Standard Class DP-13000

2 2 US-2023-112326 Date/Time 2/2023 Standard Class PO-13000

3 3 US-2023-112326 Date 2/2023 Standard Class PO-13000

4 4 US-2023-112326 Time 2/2023 Standard Class PO-13000

5 5 US-2023-141817 Date/Time/Timezone 2/2023 Standard Class MB-13000

6 6 US-2023-167199 Duration 2/2023 Standard Class ME-13000

7 7 US-2023-167199 Text 2/2023 Standard Class ME-13000

8 8 US-2023-106054 True/False 2/2023 First Class JO-13000

9 9 US-2023-167199 Binary 2/2023 Standard Class ME-13000

10 10 US-2023-167199 Using Locale... 2/2023 Standard Class ME-13000

11 11 US-2023-167199 2/2023 Standard Class ME-13000

12 12 US-2023-130813 1/6/2023 1/8/2023 Second Class LS-17

13 13 US-2023-167199 1/6/2023 1/10/2023 Standard Class ME-13000

14 14 US-2023-167199 1/6/2023 1/10/2023 Standard Class ME-13000

15 15 US-2023-105417 1/7/2023 1/12/2023 Standard Class VS-21

16 16 US-2023-105417 1/7/2023 1/12/2023 Standard Class VS-21

17 17 US-2023-135405 1/9/2023 1/13/2023 Standard Class MS-13000

18 18 US-2023-135405 1/9/2023 1/13/2023 Standard Class MS-13000

19 19 US-2023-149020 1/10/2023 1/15/2023 Standard Class AJ-13000

20 20 US-2023-149020 1/10/2023 1/15/2023 Standard Class AJ-13000

Change Type with Locale

Change the data type and select the locale of origin.

Data Type

Date

Locale

English (United States)

Sample input values:

3/29/2016

Tuesday, March 29, 2016

March 29

March 2016

OK Cancel

Change Ship Date Using Locale

The same procedure was followed for the Ship Date column to convert it into the appropriate date format.

Screenshot:

TransformColumnTypes("#Changed Type", {"Order Date", type date}, "en-US")

	Order ID	Order Date	Ship Date	Ship Mode	Customer ID
%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid
%	Error 0%	Error 0%	Error 0%	Error 0%	Error
%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty
1	US-2023-103800	03-01-2023	1/7/2023	Standard Class	DP-13000
2	US-2023				
3	US-2023				
4	US-2023				
5	US-2023				
6	US-2023				
7	US-2023				
8	US-2023				
9	US-2023				
10	US-2023				
11	US-2023				
12	US-2023				
13	US-2023				
14	US-2023				
15	US-2023				
16	US-2023				
17	US-2023				
18	US-2023				
19	US-2023				
20	US-2023-149020	10-01-2023	1/15/2023	Standard Class	AJ-10780
21	US-2023-130092	11-01-2023	1/14/2023	First Class	SV-20365

Change Type with Locale

Change the data type and select the locale of origin.

Data Type
Date

Locale
English (United States)

i Sample input values:
 3/29/2016
 Tuesday, March 29, 2016
 March 29
 March 2016

OK Cancel

STEP : 4

Checking Missing Values Using Column Quality

The Column Quality feature was used to check the quality of the data. It displays the percentage of Valid, Error, and Empty values in each column.

Steps:

View → Column Quality

Screenshot:

File Home Transform Add Column View Tools Help

Query Settings: Formula Bar, Monospaced, Column distribution, Show whitespace, Column profile, Column quality, Go to Column, Always allow, Advanced Editor, Query Dependencies

Queries [1] **Table.TransformColumnTypes(#"Changed Type with Locale", {[{"Ship Date", type date}], "en-US"}**

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
1	US-2023-103800	03-01-2023	07-01-2023	Standard Class	DP-13000	Darren Power
2	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phyllina Ober
3	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phyllina Ober
4	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phyllina Ober
5	US-2023-141817	05-01-2023	12-01-2023	Standard Class	MB-18085	Mick Brown
6	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
7	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
8	US-2023-106054	06-01-2023	07-01-2023	First Class	JO-15145	Jack O'Briant
9	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
10	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
11	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
12	US-2023-130813	06-01-2023	08-01-2023	Second Class	LS-17230	Lycoris Saund
13	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
14	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
15	US-2023-105417	07-01-2023	12-01-2023	Standard Class	VS-21820	Vivek Sundar
16	US-2023-105417	07-01-2023	12-01-2023	Standard Class	VS-21820	Vivek Sundar
17	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette
18	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette
19	US-2023-149020	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jaco
20	US-2023-149020	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jaco
21	US-2023-130092	11-01-2023	14-01-2023	First Class	SV-20365	Seth Vernon
22	US-2023-157147	13-01-2023	18-01-2023	Standard Class	BD-11605	Brian Dahlen
23	US-2023-109232	13-01-2023	16-01-2023	Second Class	ND-18370	Natalie DeCh
24	US-2023-157147	13-01-2023	18-01-2023	Standard Class	BD-11605	Brian Dahlen

Query Settings: Name: samplesuperstore, All Properties, APPLIED STEPS: Source, Promoted Headers, Changed Type, Changed Type with Locale, Changed Type with Locale1

STEP : 5

Removing Duplicates

Duplicate records were checked and removed from the dataset using the Remove Duplicates option in Power Query.

Steps:

Home → Remove Rows → Remove Duplicates

Screenshot:

File Home Transform Add Column View Tools Help

Close & Apply, New Source, Recent Sources, Enter Data, Data source settings, Manage Parameters, Export query results, Refresh Preview, Properties, Advanced Editor, Choose Columns, Remove Columns, Keep Rows, Remove Rows, Remove Top Rows, Remove Bottom Rows, Remove Alternate Rows, Remove Duplicates, Remove Blank Rows, Remove Empty Rows, Remove Rows containing duplicated values in the currently selected columns.

Queries [1] **Table.Distinct(#"Changed Type with Locale1", {"Row ID"})**

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name
1	US-2023-103800	03-01-2023	07-01-2023	Standard Class	DP-13000	Darren Power
2	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phyllina Ober
3	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phyllina Ober
4	US-2023-112326	04-01-2023	08-01-2023	Standard Class	PO-19195	Phyllina Ober
5	US-2023-141817	05-01-2023	12-01-2023	Standard Class	MB-18085	Mick Brown
6	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
7	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
8	US-2023-106054	06-01-2023	07-01-2023	First Class	JO-15145	Jack O'Briant
9	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
10	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
11	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
12	US-2023-130813	06-01-2023	08-01-2023	Second Class	LS-17230	Lycoris Saund
13	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
14	US-2023-167199	06-01-2023	10-01-2023	Standard Class	ME-17320	Maria Etezad
15	US-2023-105417	07-01-2023	12-01-2023	Standard Class	VS-21820	Vivek Sundar
16	US-2023-105417	07-01-2023	12-01-2023	Standard Class	VS-21820	Vivek Sundar
17	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette
18	US-2023-135405	09-01-2023	13-01-2023	Standard Class	MS-17830	Melanie Sette
19	US-2023-149020	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jaco
20	US-2023-149020	10-01-2023	15-01-2023	Standard Class	AJ-10780	Anthony Jaco
21	US-2023-130092	11-01-2023	14-01-2023	First Class	SV-20365	Seth Vernon
22	US-2023-157147	13-01-2023	18-01-2023	Standard Class	BD-11605	Brian Dahlen
23	US-2023-109232	13-01-2023	16-01-2023	Second Class	ND-18370	Natalie DeCh
24	US-2023-157147	13-01-2023	18-01-2023	Standard Class	BD-11605	Brian Dahlen

STEP : 6

Create Year Column and Name of Day

The Order Date column was used to create additional date-related columns.

To create Year:

Add Column → Date → Year → Year

To create Name of Day:

Add Column → Date → Day → Name of Day

These columns help in performing time-based analysis of the dataset.

Screenshot:

The screenshot shows the Tableau interface with the 'Add Column' menu open. The menu path is: Date > Year > Year, and Date > Day > Name of Day. The main view displays a table with columns: Row ID, Order ID, Order Date, Year, Name of Day, Customer ID, and Customer Name. The 'Year' and 'Name of Day' columns are newly created from the 'Order Date' column.

Row ID	Order ID	Order Date	Year	Name of Day	Customer ID	Customer Name
1	US-2023-103800	03-01-2023	2023	Monday	ME-17320	Maria Etezad
2	US-2023-112326	04-01-2023	2023	Tuesday	PO-19195	Phillina Ober
3	US-2023-112326	04-01-2023	2023	Tuesday	PO-19195	Phillina Ober
4	US-2023-112326	04-01-2023	2023	Tuesday	PO-19195	Phillina Ober
5	US-2023-141817	05-01-2023	2023	Wednesday	MB-18085	Mick Brown
6	US-2023-167199	06-01-2023	2023	Thursday	ME-17320	Maria Etezad
7	US-2023-167199	06-01-2023	2023	Thursday	ME-17320	Maria Etezad
8	US-2023-106054	06-01-2023	2023	Thursday	JO-15145	Jack O'Briant
9	US-2023-167199	06-01-2023	2023	Thursday	ME-17320	Maria Etezad
10	US-2023-167199	06-01-2023	2023	Thursday	ME-17320	Maria Etezad
11	US-2023-167199	06-01-2023	2023	Thursday	ME-17320	Maria Etezad
12	US-2023-130813	06-01-2023	2023	Thursday	LS-17320	Lycoris Saund
13	US-2023-167199	06-01-2023	2023	Thursday	ME-17320	Maria Etezad
14	US-2023-167199	06-01-2023	2023	Thursday	ME-17320	Maria Etezad
15	US-2023-105417	07-01-2023	2023	Friday	VS-21820	Vivek Sundar
16	US-2023-105417	07-01-2023	2023	Friday	VS-21820	Vivek Sundar
17	US-2023-135405	09-01-2023	2023	Sunday	MS-17830	Melanie Seite
18	US-2023-135405	09-01-2023	2023	Sunday	MS-17830	Melanie Seite
19	US-2023-149020	10-01-2023	2023	Monday	AJ-10780	Anthony Jaco
20	US-2023-149020	10-01-2023	2023	Monday	AJ-10780	Anthony Jaco
21	US-2023-130092	11-01-2023	2023	Tuesday	SV-20365	Seth Vernon
22	US-2023-157147	13-01-2023	2023	Thursday	BD-11605	Brian Dahlien
23	US-2023-109232	13-01-2023	2023	Thursday	ND-18370	Natalie DeCh
24	US-2023-157147	13-01-2023	2023	Thursday	BD-11605	Brian Dahlien
25	US-2023-149020	10-01-2023	2023	Monday	AJ-10780	Anthony Jaco

ype text)

☐	12 ₃ Year	☐	A ^B _C Day Name	☐
00%	Valid	100%	Valid	100%
0%	Error	0%	Error	0%
0%	Empty	0%	Empty	0%
5512		2023	Tuesday	
5.487		2023	Wednesday	
2717		2023	Wednesday	
7748		2023	Wednesday	
4.884		2023	Thursday	
4078		2023	Friday	
4796		2023	Friday	
2398		2023	Friday	
4.491		2023	Friday	
3112		2023	Friday	
0084		2023	Friday	
3312		2023	Friday	
1092		2023	Friday	
6742		2023	Friday	
7096		2023	Saturday	
2525		2023	Saturday	
1.168		2023	Monday	
9.75		2023	Monday	
2954		2023	Tuesday	
3583		2023	Tuesday	
0814		2023	Wednesday	
9294		2023	Friday	
3504		2023	Friday	
5.567		2023	Friday	
5.312		2023	Friday	

STEP : 7

Close and Apply

After completing the required data cleaning and transformation steps, the Close & Apply option was selected to apply all the changes and load the transformed dataset into Power BI.

Steps:

Home → Close & Apply

Screenshot:

